

Jinho Joo

☎ +82 10 5098 6620 | ✉ jinho381@yonsei.ac.kr | 🌐 joojinho97 | 📍 Seoul, Korea

Introduction

Ph.D. student specializing in AI-driven medical image reconstruction and generative models.
My research focuses on accelerated MRI reconstruction and quantitative MRI (qMRI) rapid mapping.
I have published work in top-tier computer vision and medical imaging venues including CVPR and MICCAI.

Education

Yonsei University

Mar. 2024 – Present

M.S.-Ph.D. Integrated Program in Electrical and Electronic Engineering

Seoul, Korea

- Research Focus: scan-specific MRI reconstruction, with emphasis on parallel MRI reconstruction and qMRI rapid mapping.
- Relevant Coursework: Principles of MRI, Deep Learning, Computer Vision, Digital Signal Processing, Optimization.

Kangwon National University

Mar. 2017 – Feb. 2024

B.S. in Electrical and Electronics Engineering & Computer Science

Chuncheon, Korea

- Completed mandatory military service during undergraduate studies (Republic of Korea Army, 2018–2019).
- Research Focus: Medical AI for biosignal reconstruction and background of Artificial intelligence.
- Relevant Coursework: Machine Learning, signal processing, Deep learning. Digital image processing.

Research & Professional Experience

Yonsei University

Mar. 2024 – Present

Graduate Researcher, M.S.-Ph.D. Integrated Program

Seoul, Korea

- Conducted research on deep learning methods for accelerated MRI reconstruction, focusing on self-supervised and scan-specific reconstruction frameworks.
- Developing rapid 3D-qMRI mapping using subject specific reconstruction framework.

Kangwon National University

Mar. 2022 – Feb. 2024

Research Intern

Chuncheon, Korea

- Conducted research on medical signal reconstruction using deep generative models, focusing on recovering multi-lead ECG signals from limited wearable sensor inputs.
- Developed deep learning pipelines for biosignal processing and reconstruction, including GAN-based architectures for physiological signal modeling.
- Participated in applied AI projects involving image-based measurement software and data-driven analysis systems.

Publications

0. AZURE: Attention-guided Zero-shot mUlti-contrast REconstruction for fast quantitative MRI

J. Joo, B. Bilgic, Y Jun and D. Hwang, *Manuscript in preparation*

- Proposed AZURE, an attention-guided zero-shot framework for multi-contrast quantitative MRI reconstruction without fully sampled reference data.

1. AeSPa: Attention-guided Self-supervised Parallel Imaging for MRI Reconstruction

J. Joo, H. Kim, H. Won, D. Lee, T. Eo, and D. Hwang, *CVPR 2025*

- Proposed a zero-shot scan-specific self-supervised MRI reconstruction framework with superior reconstruction quality and faster convergence on fastMRI knee and brain datasets.

2. ZSSPI: Zero-Shot Scan Specific Parallel Imaging MRI Reconstruction using Attention based mechanism
J. Joo, H. Kim, T. Eo, H. Won, and D. Hwang, *ISMRM 2025*
 - Presented an attention-based scan-specific MRI reconstruction approach for robust parallel imaging under undersampled acquisition settings.
3. KOMET: Kspace Optimal Masking for Efficient Training of Zero-shot MRI Reconstruction
H. Kim, **J. Joo**, D. Lee, D. Hwang, and T. Eo, *ISMRM 2025*
 - Introduced an optimal k-space masking strategy for more efficient and stable zero-shot MRI reconstruction training.
4. Twelve-Lead ECG Reconstruction from Single-Lead Signals Using Generative Adversarial Networks
J. Joo, G. Joo, Y. Kim, M.-N. Jin, J. Park, and H. Im, *MICCAI 2023*
 - Proposed a GAN-based framework for reconstructing 12-lead ECG signals from single-lead inputs for wearable healthcare applications.
5. Implementation of an Image-Based Tree Form Measurement Program
J. Joo, G. Jang, E. Cheong, and H. Im, *KIISE 2021*
 - Developed an image-based software system for automated tree form measurement to improve analysis efficiency and reduce manual effort.

Awards & Honors

Excellence Award, Software Startup Challenge Competition

- Developed a web-based streaming platform for performing arts and received the Excellence Award among competing student startup projects.

SKILLS

Languages: Korean (Native), English (Fluent)

Coding : Java, Python, C, C++, MySQL

Web Dev : HTML, CSS, JavaScript